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Document ID: PAY-021 | Classification: Internal Use

Error code 4001: Invalid payment method. Verify card details and expiry. Support 3DS if required.

Error 5003 indicates gateway timeout. Typical causes: network issues, gateway overload, DNS failure.

Payment user authentication: 3DS2 flows, biometric verification, and risk-based authentication.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

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Error code 4002: Insufficient funds. Customer must use alternative payment method or add funds.

Connection pooling for payment DB: set max_connections=50, connection_timeout=30s.

Payment transaction logging: structured logs, trace IDs, and observability for incident investigation.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
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EU payment failures: Verify PSD2 compliance, SCA completion, and issuer 3DS configuration.

For INR to USD discrepancy: verify FX rate source, conversion timestamp, and rounding rules.

Payment service deployment: blue-green releases, canary deployments, and rollback procedures.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging.
For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

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- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
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Intermittent payment failures: Review gateway logs, check rate limits, verify certificate expiry.

Chargeback reason code 10.4: customer disputes transaction. Gather proof of delivery and authorization.

Billing and invoicing: subscription management, proration, tax calculation, and invoice generation.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

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Currency mismatch: Ensure amount and currency code match. INR to USD conversion requires FX rates.

Payment API rate limit: 100 requests/second. Use exponential backoff for 429 responses.

Refund and chargeback processing: initiation, status tracking, dispute resolution, and reason codes.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.